Corporate Finance

Lecture 8 extras: Evaluating a project, reading a bond headline

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How projects are evaluated

How a project is evaluated

- A project costs I today and pays a cash flow c at the end of each of the next N years. Its **cost of capital** r is the return investors could earn elsewhere at the same risk (Lecture 6).
- **Way 1: compute a return, then compare it with the cost of capital.** Ask what return the project itself earns: the rate that makes the NPV zero. That rate is the **internal rate of return** (IRR) :

$$\text{NPV} = -I + \frac{c}{1+\text{IRR}} + \frac{c}{(1+\text{IRR})^2} + \cdots + \frac{c}{(1+\text{IRR})^N} = 0$$

$$I = \frac{c}{1+\text{IRR}} + \frac{c}{(1+\text{IRR})^2} + \cdots + \frac{c}{(1+\text{IRR})^N}$$

- Solve for IRR. Accept the project when the IRR is above the cost of capital: the project earns more than the money costs. Here the test is a rate against a rate.

How a project is evaluated (con't)

- **Way 2: discount, then compare amounts.** Do not compute the project's own return. Instead, discount every c at the cost of capital r and add up:

$$\text{PV} = \frac{c}{1+r} + \frac{c}{(1+r)^2} + \cdots + \frac{c}{(1+r)^N} \qquad \text{NPV} = \text{PV} - I$$

- Accept when the NPV is positive: the project is worth more than it costs.
- Here the **discount rate** is the cost of capital. Once it has been used, there is nothing left to compare it with: the test is an amount against an amount, PV against I.

Two equivalent approaches: a return above the cost of capital is exactly what makes the discounted cash flows exceed the investment. The two analyses give the same answer.

Bond yields

In the news this morning



Dette : le taux français dépasse le cap des 5 % pour la première fois depuis 2002 PREMIUM

Dans un contexte obligataire déjà tendu, les émeutes de la semaine dernière ont jeté de l'huile sur le feu et fait flamber la dette française. La prime de risque de la France a elle aussi bondi, s'approchant de son pic.

- Riots last week, in a tense market: investors worry about French debt.
- French bonds **fell** in price, **yields rose**: the 10-year went from **3.35%** a year ago to **above 5%**, a first since 2002.
- For a bond, a lower price **is** a higher yield. To see why: **face value**, **coupon**, **price** and **yield**.

What is a bond?

A bond is a loan cut into small pieces and sold to investors. France borrows by selling bonds; so do companies via corporate bonds. The contract fixes:

- **Face value** (or principal): the amount repaid at the end. Say 100 euros.
- **Coupon rate**: the interest paid every year, as a percentage of the face value. A 3.35% coupon (France's rate a year ago) on 100 of face value pays $3.35\% \times 100 = 3.35$ euros a year.
- **Maturity**: when the face value is repaid. Say 10 years.

Two more words are **not** in the contract. They come from the market:

- **Price**: what an investor pays today to own the bond. It changes every minute, as investors buy and sell.
- **Yield** (yield to maturity): the return an investor earns if they pay today's price and hold the bond until it is repaid.

The cash flows of the bond

Face value 100, coupon 3.35%, 10 years. The holder receives $3.35\% \times 100 = 3.35$ every year, and 100 back with the last coupon:

Year	1	2	3	4	5	6	7	8	9	10
Coupon	3.35	3.35	3.35	3.35	3.35	3.35	3.35	3.35	3.35	3.35
Face value										100
Total	3.35	3.35	3.35	3.35	3.35	3.35	3.35	3.35	3.35	**103.35**

Added up, the bond pays $10 \times 3.35 + 100 = 133.5$ euros over its life.

These cash flows are **fixed by the contract**. Riots, budgets and investor moods do not change a single one of them. The only thing that moves is the **price** people are willing to pay today for this fixed stream.

From the price to the yield

The price is what you pay today; the cash flows are what you get. The yield is the discount rate that makes the two sides equal:

$$\text{Price} = \frac{3.35}{1+y} + \frac{3.35}{(1+y)^2} + \cdots + \frac{3.35}{(1+y)^9} + \frac{103.35}{(1+y)^{10}}$$

- Same logic as the IRR of a project (first slides of this deck): the investment is the price, the project is the bond, and y is the return it earns.
- The market sets the price; the yield is then **computed** from it. Nobody sets a yield directly.
- If the price is exactly 100, the yield is exactly 3.35%: you lend 100, you receive 3.35 a year on it, you get your 100 back. Yield = coupon rate. That was the situation a year ago, when the bond was issued.

Price falls, yield rises

A year of sell-offs: the price fell from 100 to about **86.9**.

- The buyer at 86.9 still receives 3.35 a year and 100 at the end: the same 133.5 euros of cash flows for 13.1 euros less.
- Two sources of return instead of one: the coupon of 3.35 and the price discount.
- Solving the price equation with Price $= 86.9$ gives $y \approx 5.05\%$.

Price paid today	Cash flows received	Yield
100	3.35 a year + 100	3.35%
96	3.35 a year + 100	3.85%
86.9	3.35 a year + 100	5.05%

Fixed cash flows, lower price, higher return. Price and yield move in **opposite directions**: "yields rose" and "bonds fell" say the same thing.

So what does the headline mean?

- Investors fear France's debt will not be brought under control, and last week's riots made lending to France look even riskier.
- For the same cash flows, they now demand a higher return. The only way to get it is to pay **less**: they sell, the price drops, the yield climbs.
- **Who loses today:** anyone who bought French bonds a year ago. Their bond pays the same 3.35 a year, but is worth about 86.9 instead of 100 if sold now: a loss of 13% in a year.
- **Who pays tomorrow:** France. The next bond it issues must offer more than 5% to find buyers, instead of 3.35%. Higher yield $=$ higher cost of debt for the borrower, which deepens the budget problem.